Ge 158 Physical Principles of Remote Sensing
Project, Fall 2026

Objectives

Broad objectives

The term project is intended to provide you with the opportunity to apply the material covered in this course to help address a problem of interest to you. As such, the project requirements, outlined below, are broadly defined and allow plenty of room for creativity. Our expectations are that everyone puts approximately the same amount of effort into their project and makes roughly the same amount of progress from their respective starting point. We recognize that everyone came to the class with different levels of understanding of electromagnetism, remote sensing, and specific project areas, and what we want to see is that everyone took something away from the class, no matter their current skill level, and worked to improve their understanding of the physics of remote sensing.

Learning objectives

1. Reinforce your understanding of the physical principles of remote sensing.
2. Develop and refine effective scientific and technical communication skills, both written and verbal.

Project guidelines

- We are open to virtually any project that uses and builds upon the material we discussed in class while satisfying the learning objectives.
- Projects can be scientific or engineering focused.
- While creativity is encouraged, your project need not be novel or cutting-edge research.
- Your project can be related to your research.
- This class is intended to teach you the physics of remote sensing, not mathematics or data processing skills. The project requirements reflect this focus, and there are no requirements or expectations to derive complex theory or process data.

Collaborative options

You are strongly encouraged to work with a partner; please talk to Brent if you wish to work alone.

Requirements

The project is meant to be developed in two stages throughout the quarter.

- **Title and abstract** **(due by 5 pm on Oct 30; submit via email to Brent)**: The purpose of the abstract is for you to start developing and organizing your ideas. It is also an opportunity for us to assess the feasibility and applicability of your project. The abstract should contain:
    - A descriptive title that is comprehensible to a general scientific audience
    - One paragraph (~250 words) describing your question, why it's important and interesting, how you plan to use the course material to address it, and a general statement of your approach. The abstract should *explicitly* describe how your project satisfies the learning objectives listed above.
    - Include a brief statement of how you used AI tools.

  Your grade for the abstract will reflect how effectively you communicate, the extent to which you fulfilled the requirements listed above for the title and abstract, and the potential for the project to further develop your understanding of the physics of remote sensing. It is okay at this point for your

project to be vague and/or overly ambitious; modifications to your project and abstract are normal as you develop your ideas. See the grading rubric at the end of this document.

- **Final report and presentation (presentations will take place Dec 1 in class; submit report and slides via email to Brent by 5 pm on Dec 1)**: A final report of your project will take the form of a short scientific paper 4–6 pages long including figures but not including references. It should contain the following:
    - Descriptive title
    - Abstract
    - Introduction
    - Methods
    - Results and Discussion (these can be a single section or two separate sections)
    - Conclusions, including an *explicit* discussion of how your project satisfied the learning objectives listed above. Can contain discussion of (hypothetical) future work.
    - References
    - Include a brief statement of how you used AI tools.
    - A final presentation for the class. Presentations will be 10 minutes each with 5 minutes of questions afterward.

  Your grade for the final project will reflect how effectively you communicate and demonstrate your knowledge of the course material. See the grading rubric at the end of this document.

Definition of a page of text

Text pages should have 1” margins, 1.5 spacing, and 12-point font.

Grading rubric

The rubric below describes how the abstract and the final report and presentation will be graded. Weights are percentages of each component's grade.

Abstract (25% of course grade)

Criterion	Weight	What strong work looks like
Question and motivation	30%	A clear, specific question; says why it matters to a general scientific audience
Connection to course physics	40%	Names the physical principles involved (e.g., dielectric contrast, volume scattering, emission) and how they apply
Plan and feasibility	20%	A plausible approach for the time available; sources or tools identified
Writing	10%	Concise and well organized; the title explains the topic to a non-specialist

Final report and presentation (45% of course grade)

Criterion	Weight	What strong work looks like
Physical understanding	35%	Correct physics, explained in your own words; quantitative where appropriate (e.g., an order-of-magnitude estimate or a simple forward calculation)
Approach and analysis	20%	Methods suited to the question; results interpreted critically, with limitations and assumptions stated
Written report	20%	Follows the required structure; figures are clear and referenced; within the page limit; references cited
Presentation	10%	Clear and well paced; fits in 10 minutes; figures readable
Q&A defense	15%	Each partner can explain the physics and answer follow-up questions individually (graded per student)

Supplement: Example project ideas

These are starting points, not assignments. Each one can be scoped down to a single physical question. A well-chosen narrow question is better than a broad survey. The lectures listed after each idea are the ones it draws on most.

Permafrost thaw and the Batagaika megaslump (Siberia). This thaw slump in Siberia has grown for decades as ground ice melts and the headwall retreats. How do frozen and thawed ground differ in dielectric properties, and how would those differences appear in radar or thermal observations? You could also look at how repeat optical or SAR imagery tracks the retreating headwall. *Lectures 1, 7, 13, 14.*

Oil spills on the ocean (e.g., the 2020 *Wakashio* spill off Mauritius). Oil changes radar returns in two ways. A thicker slick acts as a dielectric layer over seawater. A thin film damps the short capillary waves responsible for most radar backscatter, so slicks appear dark in SAR. Which effect dominates, and for what slick thicknesses and radar frequencies? Can thickness be estimated? *Lectures 1, 6, 8.*

Wildfires. Fires show up in thermal emission, smoke aerosols and burn scars. You could ask how active fires are detected from their thermal emission at different wavelengths, or how smoke affects observations at visible versus microwave frequencies. Another option is how a burn scar changes surface emissivity and radar backscatter. *Lectures 7, 9, 12, 13.*

Surface hydrology on the Greenland Ice Sheet. Each summer, meltwater forms lakes and streams on the ice surface and can drain rapidly to the bed. How deep are surface lakes, judging from how light attenuates through the water? Or how does wet snow change radar backscatter enough to map where the surface is melting? *Lectures 1, 2, 9, 13.*

Surface hydrology on Antarctic ice shelves. Meltwater on ice shelves may help drive shelf breakup. Some of it sits on the surface, and some is buried under snow or refrozen ice. How can lakes and slush be told apart from dry firn? Could radar detect water buried beneath a snow layer? *Lectures 1, 6, 9.*

Mapping a glacier's equilibrium line altitude. The equilibrium line separates the area where a glacier gains mass from the area where it loses mass. Above it there is dry or wet firn, and below it bare ice. How do these surfaces differ in optical albedo, radar volume scattering and surface roughness? Which observations best locate the line at the end of the melt season? *Lectures 7, 9, 13.*

Crevassing and damage on Antarctic ice shelves. Crevasses weaken ice shelves and can help them break up. How do crevasse fields change radar backscatter through surface roughness and their geometry relative to the radar's look direction? How might that be used to map how damaged an ice shelf is, and what are the limits of the method? *Lectures 7, 14.*

Glacier dynamics. Glacier flow speeds are routinely measured from space, by tracking features between images or by SAR interferometry. What physics makes each method work? Where does each fail, for example where the surface changes through melt or snowfall? *Lectures 7, 14.*

Wind speed over the ocean. Wind raises short capillary waves, which control radar backscatter, and roughens the surface, which changes its microwave emission. How do radar scatterometers and microwave radiometers infer wind speed, and why does the measurement depend on viewing angle and polarization? *Lectures 5, 8, 13.*

Rainwater runoff in cities. Concrete and pavement keep water from soaking in, which changes how water moves through cities. How do soil moisture, standing water and impermeable surfaces differ in dielectric properties and roughness? How would those differences appear in radar or thermal observations of an urban area? Flooded streets can be either very dark or very bright in SAR, depending on the geometry. *Lectures 1, 5, 7, 13.*

Shoreline erosion. Coastlines retreat through storms, sea-level rise and changes in sediment supply. What physical contrasts let us find the waterline in optical and radar imagery? How do tides, waves and wet sand complicate the measurement? *Lectures 1, 5, 7.*